QPCODE: 25900819

MAHATMAGANDHIUNIVERSITY,KOTTAYAM
MGU-UGP(HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

1st SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCEVS100 – ENVIRONMENTAL BIOLOGY

(2024 ADMISSION ONWARDS)

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)
Studentsshouldattemptatleastonequestionfromeachcourseoutcometoenhancetheir overall Outcome attainability

Duration:1.5 Hour Max.Marks:50

Part A - Short Answer Questions
Answer any 10 Questions - Each Question carries 1 mark

1	Who first coined the term "ecosystem"?		[K] / [CO1]
		The term "ecosystem" was coined by British ecologist Arthur Tansley in 1935.	
2	Define the term "community" in ecology?		[U] / [CO1]
		In ecology, a community is a group of two or more different species' populations that coexist and interact in the same geographical area at the same time.	
3	What is the importance of biodiversity?		[An] / [CO3]
		Biodiversity is important because it provides essential ecosystem services like clean air, water, and food, maintains ecological balance, and makes ecosystems more resilient to environmental changes.	
4	What type of root system is typical in mangrove trees?		[K] / [CO2]
		Mangrove trees typically have a complex root system that includes both underground anchor roots (like flat, cable, or heart roots) and aerial roots (also called "breathing roots" or pneumatophores), which are specialized for gas exchange in oxygen-poor soil. These aerial roots come in various forms, such as stilt roots, knee roots, and pencil-like roots, depending on the species.	
5	How has the concept of homeostasis is changed over time		[U] / [CO2]
		The concept of homeostasis has evolved from a simple, static model of maintaining internal stability to a more dynamic view that incorporates behavioral responses, adaptation to changing set points, and application to ecological and evolutionary systems.	
6	Why is nitrogen important for life		[U] / [CO2]
		Nitrogen is crucial for life because it is a fundamental building block of proteins (via amino acids) and nucleic acids like DNA and RNA, which are essential for cell structure, function, and genetic information.	
7	What is the direction of flow of energy in ecosystem.		[U] / [CO2]
		The flow of energy in an ecosystem is unidirectional, meaning it flows in one direction from the sun to producers (plants) and then to consumers at higher trophic levels, without being recycled back	

8		How is an ecological pyramid important	[U] / [CO2]
		An ecological pyramid is important because it visualizes an ecosystem's structure, showing how energy, biomass, and numbers decrease at each trophic level, which helps us understand energy transfer efficiency and organism relationships.	
9		How does a keystone species affect an ecosystem?	[An] / [CO3]
		A keystone species significantly impacts an ecosystem by maintaining its structure and stability, preventing any single species from dominating, and supporting biodiversity.	
10		Interpret how the entrance of an alien species affects the local species diversity.	[An] / [CO5]
		The entrance of an alien species often leads to a decrease in local species diversity by outcompeting, preying upon, or disrupting native species' habitats, sometimes resulting in a decline or even extinction of native populations.	
11		Why is biodiversity important?	[U] / [CO3]
		Biodiversity is crucial because it provides essential ecosystem services like clean air, water, and food, which are necessary for human survival and well-being. A	
12		Why is biodiversity important to the planet?	[An] / [CO5]
		Biodiversity is important because it provides essential ecosystem services like clean air and water, and maintains food security, medicinal resources, and a stable climate.	
			[10x1 = 10]

Part B
Short Essay Type Questions
Answer any 6 Questions Each Question carries **5** marks

13		What is mean by hydrosphere?	[K] / [CO1]
		The hydrosphere is the total mass of water on Earth, encompassing all liquid, solid (ice), and gaseous (vapor) water found on, under, and above the surface. It includes oceans, lakes, rivers, glaciers, groundwater, and water vapor in the atmosphere, and it plays a crucial role in regulating climate and sustaining life. Surface and underground water: This includes all bodies of water like oceans, seas, lakes, and rivers, as well as groundwater and aquifers. Frozen water: This component consists of ice sheets, glaciers, and snowcover. Atmospheric water: This category includes water in the air as vapor and clouds. Key role: The hydrosphere is vital for regulating Earth's climate and maintaining the balance of the Earth system by influencing processes like the water cycle.	
14		Explain the characteristics of tundra ecosystem.	[U] / [CO2]
		Key characteristics of the tundra ecosystem: Low temperatures: Winters are long and brutally cold, while summers are short and cool, with average	

		temperatures often below freezing. Permafrost: A permanently frozen layer of soil, rock, and organic matter lies beneath the surface, preventing water from draining and limiting plant root growth. Short growing season: The growing season is brief, lasting only about 50 to 60 days, during which plants must grow, reproduce, and complete their life cycle quickly. Minimal precipitation: The tundra receives very little rainfall, similar to a desert, though it can be wet because permafrost prevents water from draining away. Low biodiversity: Due to the harsh conditions, the variety of plant and animal species is lower than in most other biomes. Treeless and simple vegetation: The lack of trees is a defining feature. Vegetation is dominated by low-lying plants such as mosses, lichens, sedges, grasses, and dwarf shrubs. Windy and high humidity: The ecosystem is often extremely windy with high humidity. *(any five characteristic – full credit)*	
15	Describe the characteristics of desert ecosystem.		[U] / [CO2]
		Characteristics of desert ecosystem: **Climate and soil** Low precipitation: Deserts receive very little rainfall, typically less than 10 inches (250 mm) per year. Extreme temperatures: They experience large temperature swings between day and night. Days can be scorching hot, while nights can become very cold. High winds: High wind velocity is common, which can contribute to erosion and sand dunes. Low humidity: The air holds very little moisture, which is why temperature regulations are so extreme. Soil: The soil is often sandy or rocky and can lack essential organic nutrients. **Vegetation and animal life** Sparse vegetation: Due to the scarcity of water, plant life is sparse and consists of drought-tolerant species like cacti, succulents, and small shrubs. Adapted organisms: Animals and plants have evolved specific adaptations to cope with the harsh conditions. Examples include: Water conservation: Animals develop physical traits to conserve water and plants have mechanisms like deep root systems or waxy coatings. Behavioral adaptations: Many animals are nocturnal to avoid the heat of the day. Biodiversity: While life is challenging, deserts support a variety of life forms, including insects, reptiles, and mammals, each with specialized survival traits. *(any five characteristic – full credit)*	
16	How does the phosphorous cycle work		[K] / [CO2]

		Steps in the phosphorus cycle: Weathering: Phosphorus starts in rocks. Weathering and erosion, often aided by rain or acid from microbes, break down these rocks and release phosphates into the soil and water. Absorption: Plants absorb inorganic phosphate from the soil through their roots. This phosphorus is used to create organic compounds like DNA. Consumption: Animals obtain phosphorus by eating plants or other animals, moving it up the food chain. Decomposition: When plants and animals die, decomposers (like bacteria and fungi) break down their organic matter. This process returns inorganic phosphate to the soil. Sedimentation: Phosphate in water can be washed into the ocean. Over millions of years, it sinks and can be compressed to form new sedimentary rocks, which completes the cycle.	
17	How do we measure the productivity of ecosystem		[U] / [CO2]
		Ecosystem productivity is measured by the rate of biomass or organic matter produced per unit area over time, typically expressed as grams per square meter per year or kilocalories per square meter per year. Different methods are used, such as the light-dark bottle technique for aquatic systems to measure oxygen changes, and techniques like carbon dioxide flux measurements and remote sensing for terrestrial ecosystems. **Measurement methods** **For aquatic ecosystems:** Light-Dark Bottle Technique: Paired bottles are placed in the water. A clear bottle measures the change in oxygen from both photosynthesis and respiration, while a dark bottle measures only respiration. The difference indicates net primary productivity. Carbon-14 uptake: **For terrestrial ecosystems:** Carbon Dioxide Flux: A clear, sealed chamber ("tent") is placed over a community. The change in carbon dioxide concentration in the air is measured to determine the net effect of photosynthesis and respiration. Biomass sampling: Directly measuring the dry weight of organic matter in a given area over a specific time period. Remote sensing:	
18	Discuss the scope of biodiversity conservation.		[U] / [CO3]
		Scope of biodiversity conservation Preserving species diversity: Protecting different animal, plant, and other organisms to prevent extinction and maintain a wide variety of life on Earth. Maintaining ecosystem sustainability: Ensuring the long-term health of ecosystems to maintain essential life-supporting processes like the water cycle and pollination. Safeguarding genetic diversity: Protecting the genetic variations within and between species, which is vital for agriculture, crop breeding, and a species' ability to adapt to environmental changes. Sustainable resource use: Promoting the responsible use of biological resources to ensure they are available for future generations. Protection of traditional knowledge: Integrating and protecting the traditional ecological knowledge of local communities alongside the management of biological resources.	
19		Describe the causes of biodiversity depletion.	[U] / [CO5]
		a. Habitat loss and fragmentation - Deforestation, urbanization, and agricultural expansion: destroy natural environments that species depend on for survival. This process often fragments habitats,	

		breaking them into smaller, isolated pieces that reduce available living space and affect species' ability to find food, reproduce, and thrive. b. Climate change - Alterations in temperature and precipitation patterns directly impact habitats and the life cycles of many species.It can force species to migrate, adapt, or face extinction if they cannot cope with the changes. c. Pollution - Agricultural, industrial, and waste-based pollutants: contaminate ecosystems through chemical runoff, plastic debris, and other contaminants. Pollution can directly harm wildlife, degrade habitats, and disrupt the reproductive systems of sensitive species. d. Overexploitation of resources - Activities such as overfishing, hunting, and poaching directly reduce the populations of certain species. Over-harvesting and over-consumption of natural resources contribute to a decline in biodiversity. e. Invasive species - The introduction of non-native, or "invasive," species into new ecosystems can outcompete native species for resources. These new species can also alter the structure of the native habitat, further threatening native wildlife.	
20		Point out the characteristics of grasslands	[U] / [CO2]
		Climate Semi-arid conditions: Grasslands receive more rainfall than deserts but less than forests, typically between 250 and 950 mm (10 to 37 inches) per year. Seasonal variation: Temperate grasslands have cold winters and warm summers, while tropical grasslands (savannas) have hot and wet seasons alternating with hot and dry seasons. Prone to drought: The climate is often prone to periodic droughts and strong winds. **Vegetation** Grasses dominate: The defining characteristic is the dominance of grasses and other herbaceous plants. Few trees: There are few trees, which are often limited to areas near rivers or streams. Wildflowers: Wildflowers and shrubs are also common, especially during certain seasons. Fire adaptations: Grasses are well-adapted to survive periodic wildfires, with deep roots that allow them to regrow quickly after a fire. Soil Deep and fertile: The soil is often deep, dark, and rich in nutrients from the decomposition of dead plant matter, making it ideal for farming. Nutrient-dense: The constant cycle of grass dying back in winter or the dry season and regrowing contributes to the soil's fertility. **Wildlife** Large herbivores: Grasslands support large grazing animals like bison, zebras, and elephants. Predators: Carnivores such as lions, wolves, and cheetahs are also found in these ecosystems. Small animals and birds: They are also home to numerous small mammals, such as prairie dogs and rabbits, as well as various insects and birds. Burrowing animals: Many grassland animals have adapted to the environment by burrowing to escape predators, harsh weather, and fire. *(any five relevant characteristic – full credit)*	
			(5 × 6 = 30)

	Part C Essay Type Questions Answer any **1** Question Question carries **10** marks		
21	What are the different types of aquatic ecosystems?		[U] / [CO1]
		Aquatic ecosystems are broadly classified into freshwater, marine, and brackish water types based on salinity. Freshwater ecosystems include still waters like lakes and ponds (lentic) and flowing waters like rivers and streams (lotic). Marine ecosystems are salty oceans and seas, while brackish ecosystems are where fresh and saltwater mix, such as estuaries and mangroves. **Freshwater ecosystems** Lentic: Still or slow-moving water bodies. Examples: Lakes, ponds, and some wetlands. Lotic: Flowing or fast-moving water. Examples: Rivers, streams, and creeks. Wetlands: Areas saturated with water, like marshes, swamps, and bogs. **Marine ecosystems** Salty water found in oceans, seas, and coastal areas. Examples: Open ocean, coral reefs, and coastal zones. **Brackish water ecosystems** A mix of freshwater and saltwater. Examples: Estuaries, which are where rivers meet the sea, and mangrove forests.	
22	Describe Red data book. Explain its relevance.		[U] / [CO3]
		A Red Data Book is a comprehensive public document that records and catalogs rare and endangered species of plants, animals, and fungi. Its main purpose is to identify and safeguard species on the brink of extinction to inform research and conservation efforts. The IUCN Red List is the most well-known global version of this, which helps raise awareness and guides conservation strategies by providing data on species' status and risks. **Relevance of the Red Data Book** Biodiversity conservation: It is crucial for biodiversity conservation as it provides valuable information that can be used to develop policies and strategies for protecting vulnerable species. Species identification: It helps identify which species are rare, at risk, or on the verge of extinction, serving as a crucial resource for monitoring and research. Conservation efforts: By documenting endangered species, it helps conservationists and researchers take necessary actions to protect them. Research and monitoring: It provides comprehensive data for research, studies, and monitoring programs focused on rare and endangered species and their habitats. International cooperation: It facilitates global cooperation in conservation efforts by providing a common framework for assessing species risk and highlighting the need for concerted action. Status assessment: The Red Data Book helps in evaluating the conservation status and risk of endangerment for various species on a global scale. Documentation of extinct species: The book includes a section with black pages to list species that are already extinct, serving as a record of species that have been lost forever.	
			(10 × 1 = 10)